

Problem Set 3

Applied Stats/Quant Methods 1

Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in R using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
1 model_bivar_1 <- lm(voteshare ~ difflog, data = inc.sub)
2 summary(model_bivar_1)
```

A bivariate regression analysis was run by using the `lm` function in R. A bivariate regression was run because there was an outcome variable, `voteshare` and one explanatory variable, `difflog`

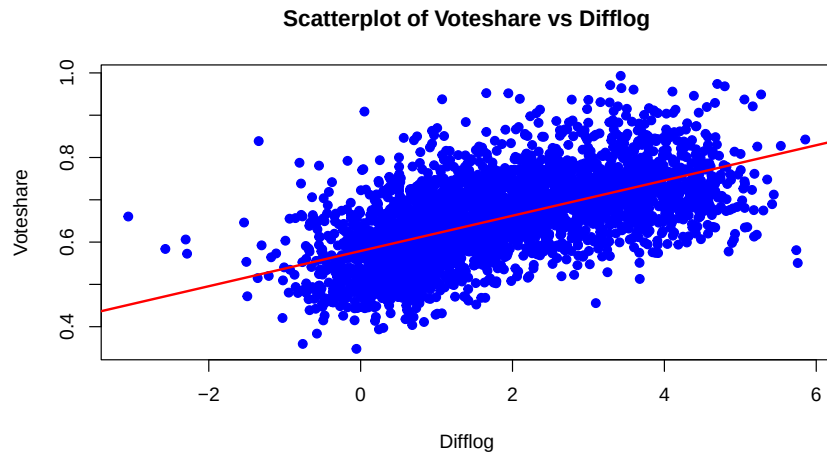
2. Make a scatterplot of the two variables and add the regression line.

```

1 plot(inc.sub$difflog, inc.sub$voteshare,
2       xlab = "Difflog", ylab = "Voteshare",
3       main = "Scatterplot of Voteshare vs Difflog",
4       pch = 19, col = "blue")
5 abline(lm(voteshare ~ difflog, data = inc.sub), col = "red", lwd = 2)

```

I created a scatter plot of the two variables that I would like to observe and also created a regression line through the scatter plot.



3. Save the residuals of the model in a separate object.

```

1 residuals_model_1 <- residuals(model_bivar_1)
2 head(residuals_model_1)

```

4. Write the prediction equation.

$$\widehat{voteshare}_i = \beta_0 + \beta_1 difflog_i$$

I used Latex code to do this.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

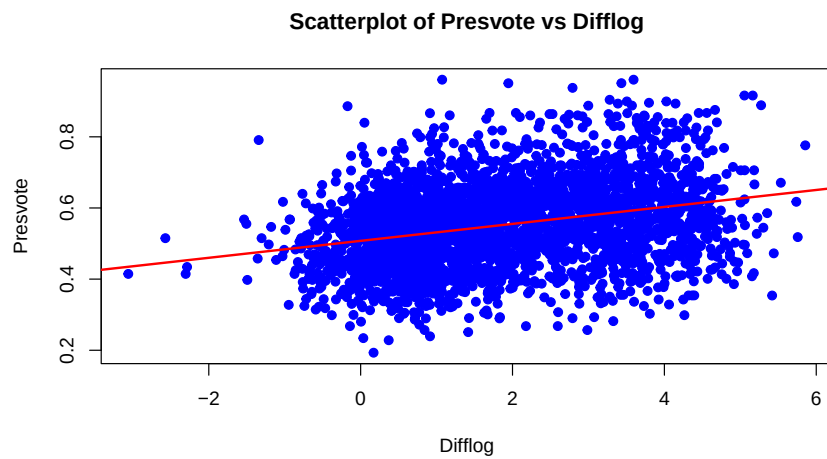
```
1 model_bivar_2 <- lm(presvote ~ difflog, data = inc.sub)
2 summary(model_bivar_2)
```

A bivariate regression analysis was run by using the `lm` function in R. A bivariate regression was run because there was an outcome variable, `presvote` and one explanatory variable, `difflog`

2. Make a scatterplot of the two variables and add the regression line.

```
1 plot(inc.sub$difflog, inc.sub$voteshare,
2      xlab = "Difflog", ylab = "Presvote",
3      main = "Scatterplot of Presvote vs Difflog",
4      pch = 19, col = "blue")
5 abline(lm(presvote ~ difflog, data = inc.sub), col = "red", lwd = 2)
```

I created a scatter plot of the two variables that I would like to observe and also created a regression line through the scatter plot.



3. Save the residuals of the model in a separate object.

```
1 residuals_model_1 <- residuals(model_bivar_2)
2 head(residuals_model_1)
```

4. Write the prediction equation.

$$\widehat{presvote}_i = \beta_0 + \beta_1 difflog_i$$

I used Latex code to do this.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is **voteshare** and the explanatory variable is **presvote**.

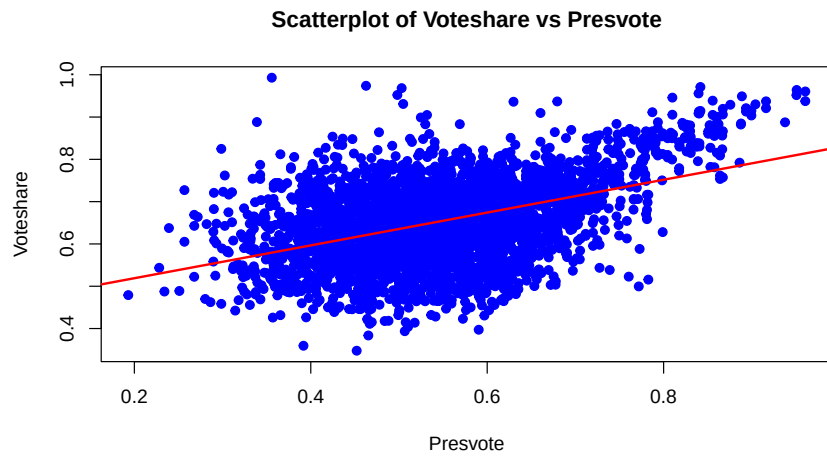
```
1 model_bivar_3 <- lm(voteshare ~ presvote, data=inc.sub)
2 summary(model_bivar_3)
```

A bivariate regression analysis was run by using the lm function in R. A bivariate regression was run because there was an outcome variable, voteshare and one explanatory variable, presvote

2. Make a scatterplot of the two variables and add the regression line.

```
1 plot(inc.sub$presvote, inc.sub$voteshare,
2      xlab = "Presvote",
3      ylab = "Voteshare",
4      main = "Scatterplot of Voteshare vs Presvote",
5      pch=19, col="blue")
6 model <- lm(voteshare ~ presvote, data=inc.sub)
7 abline(model, col="red", lwd=2)
```

I created a scatter plot of the two variables that I would like to observe and also created a regression line through the scatter plot.



3. Write the prediction equation.

$$\widehat{voteshare}_i = \beta_0 + \beta_1 presvote_i$$

I used Latex code to do this.

Question 4

The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

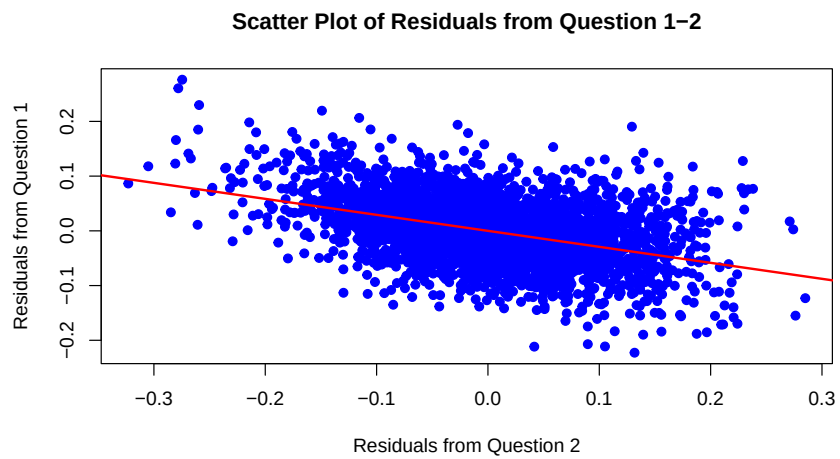
1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
1 resid_model <- lm(residuals_model_1 ~ residuals_model_2)
2 summary(resid_model)
```

A bivariate regression analysis was run by using the lm function in R. A bivariate regression was run because there was an outcome variable, residuals from question 1 and one explanatory variable, the residuals from Question 2.

2. Make a scatterplot of the two residuals and add the regression line. *I created a scatter plot of the two variables that I would like to observe and also created a regression line through the scatter plot.*

```
1 plot(residuals_model_2, residuals_model_1,
2       xlab = "Residuals from Question 2",
3       ylab = "Residuals from Question 1",
4       main = "Scatter Plot of Residuals from Question 1-2",
5       pch = 19,
6       col = "blue")
7 residual_model <- lm(residuals_model_1 ~ residuals_model_2)
8 abline(residual_model, col = "red", lwd = 2)
```



3. Write the prediction equation.

$$\text{residuals1} = \beta_0 + \beta_1 \cdot \text{residuals2}$$

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
1 model_multi_4 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
2 summary(model_multi_4)
```

This is a multivariate regression where multiple explanatory variables (`difflog` and `presvote`) are being compared against one outcome variable, vote share. I changed the regression analysis in R to reflect this.

2. Write the prediction equation. *For a multivariate analysis you include all the variables in the equation and even the ones you are not analyzing. Therefore, my prediction equation includes all the variables.*

$$\begin{aligned}\widehat{\text{voteshare}} = & \beta_0 + \beta_1 \cdot \text{difflog} + \beta_2 \cdot \text{presvote} + \beta_3 \cdot \text{unemployed} + \beta_4 \cdot \text{milpop} + \beta_5 \cdot \text{age65} \\ & + \beta_6 \cdot \text{urban} + \beta_7 \cdot \text{population} + \beta_8 \cdot \text{south} + \beta_9 \cdot \text{chalquality} + \beta_{10} \cdot \text{midterm} \\ & + \beta_{11} \cdot \text{seniority} + \beta_{12} \cdot \text{incparty} + \beta_{13} \cdot \text{inparty} + \beta_{14} \cdot \text{incspend} + \beta_{15} \cdot \text{chalspend} \\ & + \beta_{16} \cdot \text{congress} + \beta_{17} \cdot \text{year} + \beta_{18} \cdot x + \beta_{19} \cdot X\end{aligned}\tag{1}$$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?